

17/04/2025



Aakash

Medical | IIT-JEE | Foundations

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CODE-A



AIM - 720

(Advanced **INTENSIVE** Mastery for 720)

MM : 720

CST-6

Time : 180 Mins.

Answers

1. (1)	37. (2)	73. (2)	109. (2)	145. (4)
2. (1)	38. (1)	74. (3)	110. (2)	146. (4)
3. (4)	39. (3)	75. (2)	111. (2)	147. (2)
4. (3)	40. (4)	76. (2)	112. (2)	148. (3)
5. (4)	41. (1)	77. (1)	113. (3)	149. (2)
6. (4)	42. (2)	78. (3)	114. (3)	150. (4)
7. (4)	43. (1)	79. (1)	115. (4)	151. (4)
8. (3)	44. (2)	80. (3)	116. (1)	152. (2)
9. (1)	45. (3)	81. (3)	117. (2)	153. (1)
10. (4)	46. (2)	82. (3)	118. (2)	154. (4)
11. (1)	47. (4)	83. (3)	119. (3)	155. (4)
12. (4)	48. (2)	84. (3)	120. (2)	156. (2)
13. (3)	49. (4)	85. (2)	121. (4)	157. (2)
14. (3)	50. (1)	86. (2)	122. (3)	158. (4)
15. (2)	51. (4)	87. (2)	123. (2)	159. (1)
16. (3)	52. (1)	88. (4)	124. (3)	160. (2)
17. (1)	53. (4)	89. (3)	125. (4)	161. (1)
18. (1)	54. (3)	90. (2)	126. (1)	162. (4)
19. (1)	55. (1)	91. (2)	127. (2)	163. (1)
20. (3)	56. (4)	92. (4)	128. (1)	164. (3)
21. (2)	57. (3)	93. (3)	129. (3)	165. (4)
22. (4)	58. (4)	94. (2)	130. (2)	166. (2)
23. (3)	59. (1)	95. (3)	131. (4)	167. (4)
24. (3)	60. (1)	96. (4)	132. (2)	168. (3)
25. (2)	61. (2)	97. (4)	133. (3)	169. (3)
26. (2)	62. (3)	98. (1)	134. (4)	170. (1)
27. (2)	63. (2)	99. (4)	135. (4)	171. (4)
28. (1)	64. (1)	100. (4)	136. (4)	172. (2)
29. (3)	65. (3)	101. (3)	137. (3)	173. (2)
30. (3)	66. (1)	102. (3)	138. (1)	174. (3)
31. (1)	67. (1)	103. (4)	139. (2)	175. (1)
32. (2)	68. (2)	104. (2)	140. (2)	176. (2)
33. (3)	69. (3)	105. (3)	141. (1)	177. (4)
34. (4)	70. (2)	106. (2)	142. (2)	178. (4)
35. (1)	71. (2)	107. (4)	143. (3)	179. (3)
36. (4)	72. (2)	108. (3)	144. (4)	180. (3)

Hints and Solutions

PHYSICS

(1) Answer : (1)

Solution:

Relative velocity of two objects

$$\vec{v}_{A/B} = \vec{v}_A - \vec{v}_B$$

$$\text{Hence } |v_A - v_B| \leq |\vec{v}_{A/B}| \leq |v_A + v_B|$$

For two cars moving in opposite directions,

$$|\vec{v}_{A/B}| = |v_A + v_B|$$

$$= 20 + 30 = 50 \text{ m/s}$$

(2) Answer : (1)

Solution:

When nucleus is formed, some mass is lost in form of energy to make it stable.

(3) Answer : (4)

Solution:

All the given types of semiconductors are electrically neutral.

(4) Answer : (3)

Solution:

$$E_{\text{required}} = 13.6 \left[\frac{1}{4} - \frac{1}{9} \right] z^2$$

$$30.24 = 1.89 z^2$$

$$z^2 = 16$$

$$z = 4$$

(5) Answer : (4)

Solution:

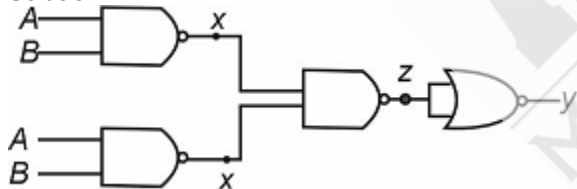
$$\text{De-Broglie wavelength } \lambda = \frac{h}{mv} = \frac{h}{\sqrt{2mqV}}$$

$$\lambda \propto \frac{1}{\sqrt{m}}$$

$$\therefore \frac{\lambda_e}{\lambda_p} = \sqrt{\frac{m_p}{m_e}}$$

(6) Answer : (4)

Solution:



$$x = \overline{A \cdot B}$$

$$z = \overline{\overline{A \cdot B} \cdot \overline{A \cdot B}}$$

$$y = \overline{\overline{\overline{A \cdot B} \cdot \overline{A \cdot B}}}$$

$$= \overline{\overline{A \cdot B} \cdot \overline{A \cdot B}}$$

$$= \overline{A \cdot B}$$

= NAND gate

(7) Answer : (4)

Solution:

$$\tau_{\text{fulcrum}} = 0$$

$$W \times 30 = 300 \times 50$$

$$\Rightarrow W = 500 \text{ N}$$

(8) Answer : (3)

Solution:

$$v_T = \frac{2}{9} \times \frac{a^2}{\eta} \times (\rho_0 - \rho_l) \times g$$

$$\Rightarrow v_T \propto a^2, \rho = \frac{M}{V} \Rightarrow \frac{M}{\frac{4}{3}\pi a^3} = \frac{M'}{\frac{4}{3}\pi (a')^3}$$

$$\Rightarrow \frac{M}{a^3} = \frac{4M}{(a')^3} \Rightarrow (a')^3 = 4a^3$$

$$\Rightarrow a' = (4)^{\frac{1}{3}} \times a$$

$$\Rightarrow \frac{v}{v'} = \frac{a^2}{(4)^{\frac{1}{3}} \times a^2} \Rightarrow v' = (16)^{\frac{1}{3}} v$$

(9) Answer : (1)

Solution:

According to equation of continuity, $\rho Av = \text{constant}$

(10) Answer : (4)

Solution:

The change in the internal energy should remain same. According to first law of thermodynamics, we know that $Q = \Delta U + W$
 $\Rightarrow \Delta U = Q - W$

(11) Answer : (1)

Solution:

$$Q_{\text{net}} = W_{\text{net}} = 200 \times 10^{-6} \times 100 \times 10^3 = -20 \text{ J}$$

(12) Answer : (4)

Solution:

$$PV = nRT$$

$$\text{Now } P' = 2P \text{ and } V' = \frac{V}{2}$$

$$\Rightarrow P'V' = PV \Rightarrow \text{temperature would remain constant.}$$

Rotational kinetic energy of an ideal monoatomic gas is zero.

(13) Answer : (3)

Solution:

The ice point in Fahrenheit scale is 32°F , while steam point in Fahrenheit scale is 212°F

(14) Answer : (3)

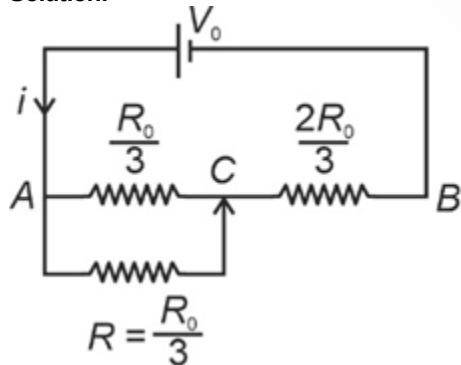
Solution:

$$\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$$

$$\Rightarrow \epsilon_0 = \frac{1}{4\pi \times 9 \times 10^9} = 8.85 \times 10^{-12} \text{ C}^2 \text{ N}^{-1} \text{ m}^{-2}$$

(15) Answer : (2)

Solution:



Equivalent resistance between points A and C is $\frac{R_0/3}{2} = \frac{R_0}{6}$

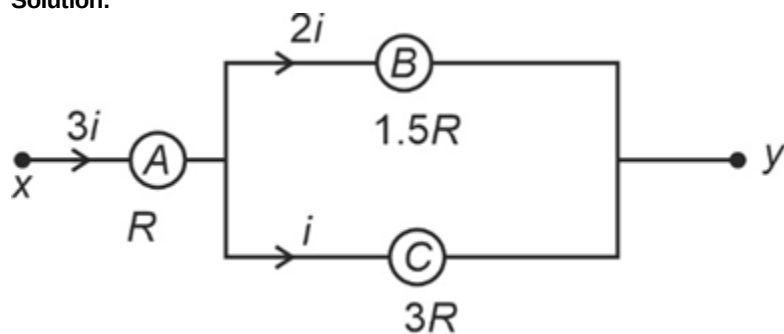
$$R_{\text{eq}} = \frac{2R_0}{3} + \frac{R_0}{6} = \frac{5R_0}{6}$$

$$i = \frac{V_0}{R_{\text{eq}}} = \frac{V_0}{\frac{5R_0}{6}} = \frac{6V_0}{5R_0}$$

$$V_{AC} = i \frac{R_0}{6} = \frac{6V_0}{5R_0} \times \frac{R_0}{6} = \frac{V_0}{5}$$

(16) Answer : (3)

Solution:



$$V_A = 3iR, V_B = 2i \times 1.5R = 3iR,$$

$$V_C = i \times 3R = 3iR$$

$$\therefore V_A = V_B = V_C$$

(17) Answer : (1)

Solution:

In a capacitor, the energy is stored in electric field between the plates

(18) Answer : (1)

Solution:

$$\phi_{\text{net}} = \frac{q_{\text{en}}}{\epsilon_0} \Rightarrow \phi_{\text{net}} = 2\phi - \phi = \frac{q}{\epsilon_0}$$

$$q = \phi \epsilon_0$$

(19) Answer : (1)

Solution:

$$\text{For a point charge : } E = \frac{kq}{r^2}$$

$$\text{For a short dipole : } V = \frac{kP \cos \theta}{r^2}$$

(20) Answer : (3)

Solution:

$$\Delta l = \frac{FL}{AY}$$

$$l = \frac{F}{\pi r^2} \frac{L}{Y} \dots (i)$$

$$l' = \frac{4FL}{\pi(4r)^2 Y} \dots (ii)$$

$$\frac{l}{l'} = 4$$

$$l' = \frac{l}{4}$$

(21) Answer : (2)

Solution:

$$y = 40 \sin \left[\frac{\pi}{5} t - \frac{\pi}{9} x \right]$$

$$Y = A \sin(\omega t - kx)$$

$$A = 40 \text{ cm}, \omega = \frac{\pi}{5} = \frac{2\pi}{T}, k = \frac{\pi}{9}$$

$$T = 10 \text{ seconds}$$

$$\frac{2\pi}{\lambda} = \frac{\pi}{9}$$

$$\lambda = 18 \text{ cm}$$

$$v = \frac{\omega}{k} = \frac{\pi/5}{\pi/9} = \frac{9}{5} = 1.8 \text{ cm/s}$$

(22) Answer : (4)

Solution:

In EM wave, electric and magnetic field components are in phase and perpendicular to each other and also perpendicular to direction of propagation.

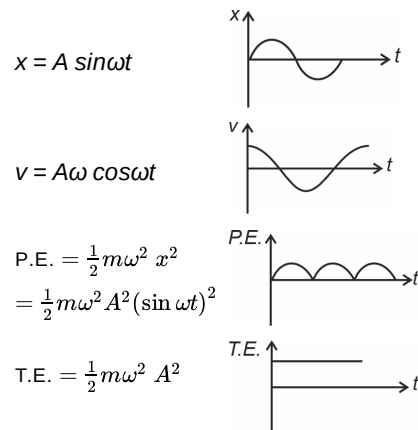
(23) Answer : (3)

Solution:

For n^{th} minima, path difference is $(2n-1)\frac{\lambda}{2}$.

For 5th minima, path difference is $= \frac{[2(5)-1]\lambda}{2} = \frac{9\lambda}{2}$

(24) Answer : (3)

Solution:


(25) Answer : (2)

Solution:

$$V \propto L^3$$

$$\% \text{ change in volume} = \frac{\Delta V}{V} \times 100 = \frac{3\Delta L}{L} \times 100$$

$$= 3 \times 4\% = 12\%$$

(26) Answer : (2)

Solution:

$$\text{Energy } [E] = [ML^2T^{-2}]$$

$$a = 1, b = 2, c = -2$$

$$x_2 = x_1 \left[\frac{M_1}{M_2} \right]^a \left[\frac{L_1}{L_2} \right]^b \left[\frac{T_1}{T_2} \right]^c$$

$$= 1 \left(\frac{\text{kg}}{100 \text{ gm}} \right)^1 \left(\frac{\text{m}}{10 \text{ cm}} \right)^2 \left(\frac{\text{s}}{5 \text{ s}} \right)^{-2}$$

$$= \left(\frac{1000 \text{ gm}}{100 \text{ gm}} \right) \left(\frac{100 \text{ cm}}{10 \text{ cm}} \right)^2 \left(\frac{1}{5} \right)^{-2}$$

$$(10)(10)^2 \left(\frac{1}{5} \right)^{-2} = 25000$$

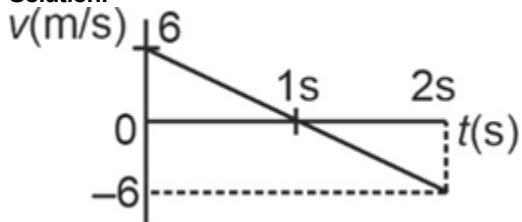
(27) Answer : (2)

Solution:

• If particle is at rest, then it will move in direction of acceleration.

• Initial direction of motion of particle may be oblique to direction of acceleration, hence parabolic path may be a possibility.

(28) Answer : (1)

Solution:


Displacement = Area under the curve = zero

$$\text{Distance} = \left(\frac{1}{2} \times 6 \times 1 \right) + \left(\frac{1}{2} \times 6 \times 1 \right) = 6 \text{ m}$$

(29) Answer : (3)

Solution:

During explosion, only internal forces are acting on the fragments of the bomb, hence total momentum will remain same as initial momentum of the bomb

(30) Answer : (3)

Solution:

$$\text{At point A : } mg - N = \frac{mv^2}{r}$$

$$1000 - 100 = \frac{mv^2}{r}$$

$$\Rightarrow \frac{mv^2}{r} = 900 \dots (i)$$

At point B :

$$N' - mg = \frac{mv^2}{r}$$

$$N' = 1000 + 900 = 1900 \text{ N [from equation (i)]}$$

(31) Answer : (1)

Solution:

$$F = - \left(\frac{\partial U}{\partial x} \hat{i} + \frac{\partial U}{\partial y} \hat{j} + \frac{\partial U}{\partial z} \hat{k} \right)$$

$$= - \left[(3x^2 - 2x) \hat{i} + (2 + 2y) \hat{j} \right]$$

$$F(1, 2) = - \left[(3 - 2) \hat{i} + (2 + 4) \hat{j} \right] = (-\hat{i} - 6\hat{j}) \text{ N}$$

(32) Answer : (2)

Solution:

$$\text{Power} = \text{Force} \times \text{speed}$$

$$2400 = 800 \times 10 \times v$$

$$v = 0.3 \text{ m/s}$$

(33) Answer : (3)

Solution:

Position of particle (A) w.r.t. particle (B) is

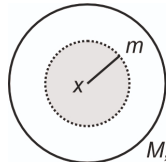
$$\vec{r}_{AB} = \vec{r}_A - \vec{r}_B = (-\hat{i} + 3\hat{j}) \text{ m}$$

$$\vec{L} = \vec{r}_{AB} \times \vec{P}_A = (-\hat{i} + 3\hat{j}) \times 4\hat{k}$$

$$\vec{L} = (12\hat{i} + 4\hat{j}) \text{ kg m}^2\text{s}^{-1}$$

(34) Answer : (4)

Solution:



The force of attraction on point mass m is given by

$$F = m \left(\frac{GMx}{R^3} \right)$$

(35) Answer : (1)

Solution:

$$T = \frac{2\pi}{\sqrt{GM_e}} R_e^{\frac{3}{2}} = 2\pi \sqrt{\frac{R_e}{g}}$$

$$\approx 84.6 \text{ minute}$$

(36) Answer : (4)

Solution:

$$I_{\text{hollow}} = I_{\text{CM}} + MR^2 = \frac{2}{3} MR^2 + MR^2 = \frac{5}{3} MR^2$$

$$I_{\text{ring}} = MR^2$$

$$I_{\text{disc}} = \frac{MR^2}{2}$$

$$I_{\text{solid}} = \frac{2}{5} MR^2$$

$$l_{\text{solid}} < l_{\text{disc}} < l_{\text{ring}} < l_{\text{hollow}}$$

(37) Answer : (2)

Solution:

According to Newton's formula

$$x_0 x_i = f^2 \Rightarrow x_i = \frac{f^2}{x_0}$$

(38) Answer : (1)

Solution:

For a real object, $u = -x_0$

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f} \Rightarrow \frac{1}{v} + \frac{1}{-x_0} = \frac{1}{f_0}$$

$$\frac{1}{v} = \frac{1}{f_0} + \frac{1}{x_0} \Rightarrow v = \frac{x_0 \times f_0}{x_0 + f_0} \dots (i)$$

$$m = \frac{-v}{u} = -\left(\frac{x_0 f_0}{x_0 + f_0}\right) \times \frac{1}{-x_0} = \frac{f_0}{x_0 + f_0}$$

$$\therefore \frac{f_0}{x_0 + f_0} < 1 \Rightarrow m < 1$$

(39) Answer : (3)

Solution:

$$\phi_{\text{net}} = \mu_0 m$$

and flux is count of number of field lines.

(40) Answer : (4)

Solution:

Inside the pipe: $B \propto r$

Outside the pipe: $B \propto \frac{1}{r}$

(41) Answer : (1)

Solution:

$$B = \frac{N\mu_0 i}{2R}$$

$$B = \frac{4\pi \times 10^{-7} \times 28 \times 5}{2 \times 20 \times 10^{-2}}$$

$$= 4 \times \frac{22 \times 10^{-5} \times 28 \times 5}{7 \times 2 \times 20}$$

$$= 4.4 \times 10^{-4} \text{ Wb/m}^2$$

(42) Answer : (2)

Solution:

$$I_{\text{rms}} = \sqrt{a^2 + \frac{b^2}{2}}$$

$$= \sqrt{(3\sqrt{2})^2 + \frac{8^2}{2}} = \sqrt{\frac{36+64}{2}}$$

$$= \frac{10}{\sqrt{2}} = 5\sqrt{2} \text{ A}$$

(43) Answer : (1)

Solution:

Resistance is independent of frequency of source.

$$X_c = \left(\frac{1}{2\pi fC}\right)$$

For d.c., $f = 0 \Rightarrow X_C = \infty$

(44) Answer : (2)

Solution:

$$q = \frac{\Delta\phi}{R} = \frac{50-10}{10} = 4 \text{ C}$$

(45) Answer : (3)

Solution:

$$U = \frac{1}{2} LI^2$$

$$8 = \frac{1}{2} \times 4 \times I^2$$

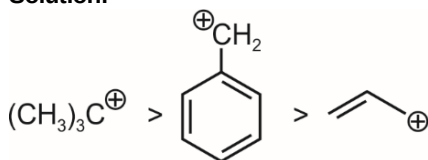
$$I^2 = 4$$

$$I = 2 \text{ A}$$

CHEMISTRY

(46) Answer : (2)

Solution:



Based on the data, the correct order of stability of the carbocations is :

(47) Answer : (4)

Solution:

Glycogen is also found in yeast and fungi

(48) Answer : (2)

Solution:

$$E = h \frac{C}{\lambda} = \frac{6.6 \times 10^{-34} \times 3 \times 10^8}{528 \times 10^{-9}}$$

$$= 0.0375 \times 10^{-17} \text{ J}$$

$$20 \text{ watt} = 20 \text{ J s}^{-1}$$

$$\text{Number of photons limited per second} = \frac{15}{0.0375 \times 10^{-17}}$$

$$= 400 \times 10^{17}$$

$$= 4 \times 10^{19}$$

(49) Answer : (4)

Solution:

More is the s character in the hybrid orbital of carbon, more is the acidic strength of the compound. + I effect causing group, decreases acidic strength of the compound.

Acidic strength order : (ii) > (iii) > (i)

(50) Answer : (1)

Solution:

Except glycine, all other naturally occurring α-amino acids are optically active

(51) Answer : (4)

Solution:

Ions		Number of unpaired electron
Mn ³⁺	→	4
Cr ⁴⁺	→	2
Ni ³⁺	→	3
Co ⁴⁺	→	5

(52) Answer : (1)

Solution:

For inversion of cane sugar rate of reaction depends on concentration of sugar only

$$r = k[\text{C}_{12}\text{H}_{22}\text{O}_{11}]$$

(53) Answer : (4)

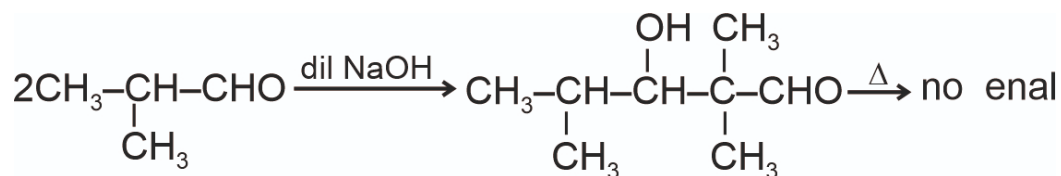
Solution:

For a zero-order reaction

$$t_{1/2} = \frac{a_0}{2k}$$

(54) Answer : (3)

Solution:



(55) Answer : (1)

Solution:

Presence of more electron withdrawing group increases the acidic strength

(56) Answer : (4)

Solution:

$[\text{Fe}(\text{SCN})]^{2+}$ is blood red in colour

(57) Answer : (3)

Solution:

Cation	Group	Reagent
Ba^{2+}	I	$(\text{NH}_4)_2\text{CO}_3 + \text{NH}_4\text{OH}$
Cu^{2+}	II	$\text{H}_2\text{S} + \text{dil. HCl}$
Al^{3+}	III	$\text{NH}_4\text{OH} + \text{NH}_4\text{Cl}$
Ni^{2+}	IV	$\text{NH}_4\text{OH} + \text{H}_2\text{S}$

(58) Answer : (4)

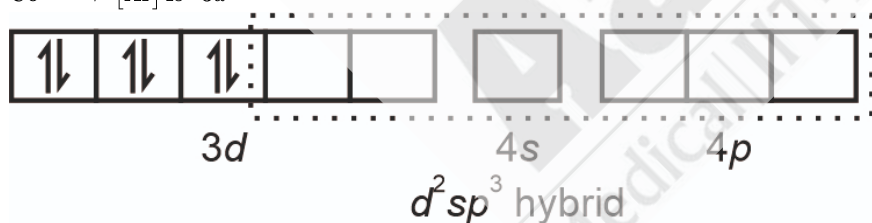
Solution:

- $[\text{Co}(\text{NH}_3)_5(\text{NO}_2)]\text{Cl}_2$ and $[\text{Co}(\text{NH}_3)_5(\text{ONO})]\text{Cl}_2$ are linkage isomers
- $[\text{Cr}(\text{NH}_3)_6][\text{Co}(\text{CN})_6]$ and $[\text{Cr}(\text{CN})_6][\text{Co}(\text{NH}_3)_6]$ are coordination isomers
- $[\text{Co}(\text{NH}_3)_5\text{SO}_4]\text{Br}$ and $[\text{Co}(\text{NH}_3)_5\text{Br}]\text{SO}_4$ are ionisation isomers
- $[\text{Cr}(\text{H}_2\text{O})_6]\text{Cl}_3$ and $[\text{Cr}(\text{H}_2\text{O})_5\text{Cl}]\text{Cl}_2 \cdot \text{H}_2\text{O}$ are solvate isomers

(59) Answer : (1)

Solution:

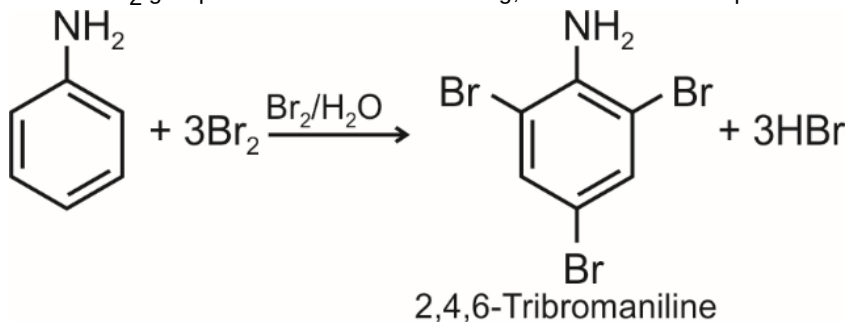
When inner 3d orbitals are used in the hybridization, formed complex is known as inner orbital complex.



(60) Answer : (1)

Solution:

When $-\text{NH}_2$ group is attached to benzene ring, it acts as ortho and para directing.

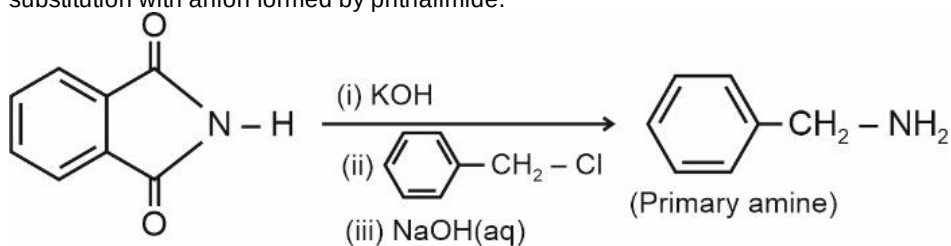


(61) Answer : (2)

Solution:

Gabriel synthesis is used for the preparation of aliphatic primary amines.

Aromatic primary amines cannot be prepared by Gabriel synthesis because aryl halides do not undergo nucleophilic substitution with anion formed by phthalimide.

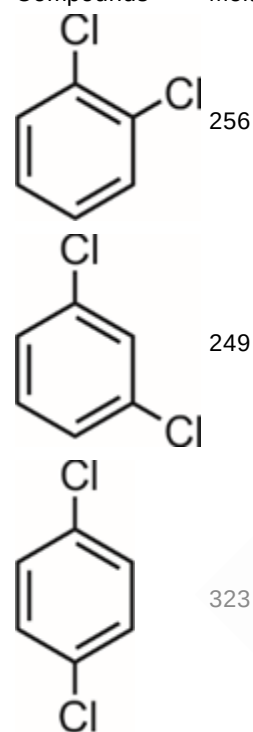


(62) Answer : (3)

Solution:

Para-isomers have high melting point than ortho and meta-isomers due to symmetry of it

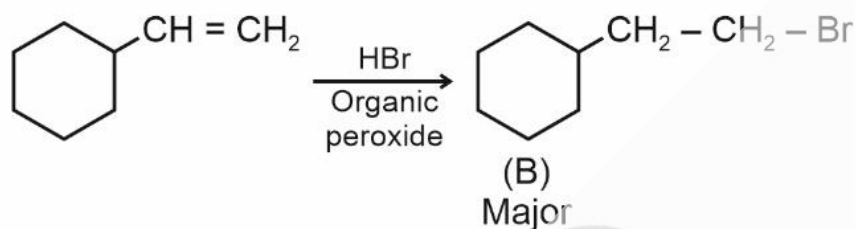
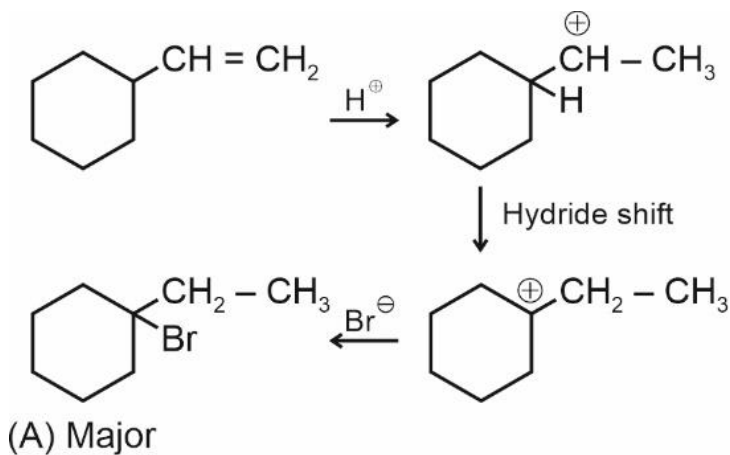
Compounds Melting point/K



(63) Answer : (2)

Solution:

In the presence of organic peroxide, addition of HBr on alkene follow anti Markovnikov's rule.



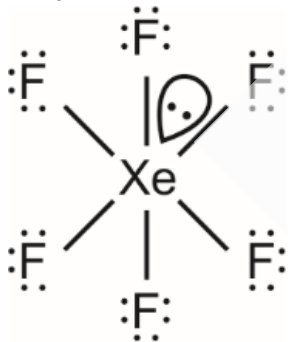
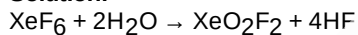
(64) Answer : (1)

Solution:

Rhombic sulphur (α -sulphur) crystal formed on evaporating the solution of roll sulphur in CS_2 .

(65) Answer : (3)

Solution:



Total number of lone pair in XeF_6 is 19.

(66) Answer : (1)

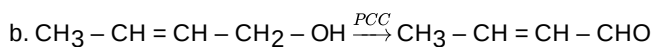
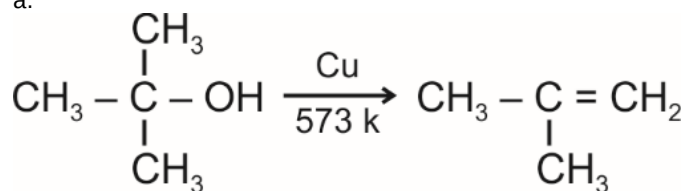
Solution:

N_2O and NO are pair of neutral oxides of nitrogen.

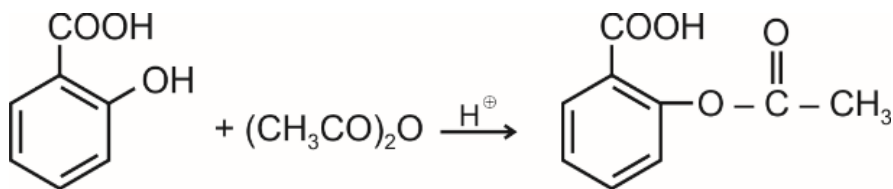
(67) Answer : (1)

Solution:

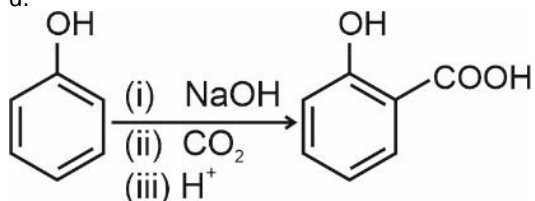
a.



c.

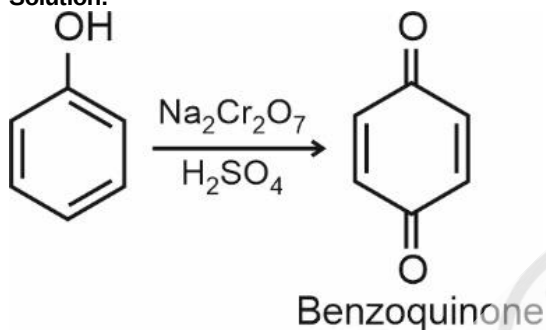


d.



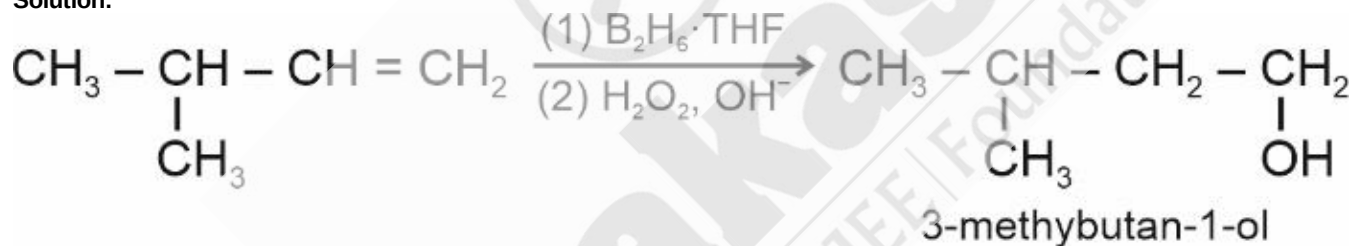
(68) Answer : (2)

Solution:



(69) Answer : (3)

Solution:



(70) Answer : (2)

Solution:

$$\text{Bond order} = \frac{1}{2} (N_b - N_a)$$

$$O_2 = \frac{1}{2} (10 - 6) = 2$$

$$O_2^- = \frac{1}{2} (10 - 7) = 1.5$$

$$N_2 = \frac{1}{2} (10 - 4) = 3$$

$$N_2^+ = \frac{1}{2} (9 - 4) = 2.5$$

(71) Answer : (2)

Solution:

There is no equilibrium between the canonical forms.

(72) Answer : (2)

Solution:

$$\text{Formal charge} = \frac{1}{2} 4 - \frac{1}{2} 8 = 0$$

(73) Answer : (2)

Solution:

1 kg of water has urea = 0.3 mole

$$= 0.3 \times 60 \text{ g} = 18 \text{ g}$$

$$\text{Total mass of solution} = 1000 + 18 = 1018 \text{ g}$$

$$3 \text{ kg of solution has urea} = \frac{18}{1.018} \times 3 = 53.04 \text{ g of urea}$$

(74) Answer : (3)

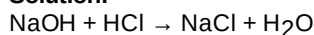
Solution:

The dissolution of gas is exothermic so on increasing temperature its solubility decreases.

The value of K_H increases with increase in temperature for O_2 .

(75) Answer : (2)

Solution:



$$\text{Molarity of NaOH} = \frac{\frac{4}{40}}{1} = 0.1 \text{ M}$$

$$\text{Moles of NaOH} = 0.1 \times 20 \times 10^{-3}$$

$$= 20 \times 10^{-4}$$

$$= 2 \times 10^{-3}$$

$$\text{Moles of HCl} = 2 \times 10^{-3}$$

$$\text{Vol of HCl} = 2 \times 10^{-3} \times 22.4$$

$$= 44.8 \times 10^{-3} \text{ L}$$

$$= 44.8 \text{ mL}$$

(76) Answer : (2)

Solution:

Ce^{4+} being $4f^0$ type and Yb^{2+} being $4f^{14}$ type species are diamagnetic in nature.

(77) Answer : (1)

Solution:

$\Delta_{hyd}H^\circ$ for Ni^{2+} is highest due to its small ionic radius and high charge density which enables it to form strong electrostatic interactions with water molecules.

Ion $\Delta_{hyd}H^\circ$ (KJ mol⁻¹)

$$Ni^{2+} \quad -2121$$

$$Co^{2+} \quad -2077$$

$$Fe^{2+} \quad -1998$$

$$Ti^{2+} \quad -1866$$

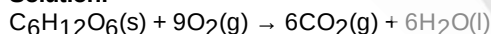
(78) Answer : (3)

Solution:

The species in its elemental form has zero standard molar enthalpy of formation at 298 K. At 298 K, Cl_2 is gas while Br_2 is liquid.

(79) Answer : (1)

Solution:



$$\Delta H_C = 6 \times \Delta H_f(CO_2) + 6 \times \Delta H_f(H_2O) - \Delta H_f(C_6H_{12}O_6)$$

$$= 6(-400) \times 6(-300) - (-1300)$$

$$= -2900 \text{ kJ/mol}$$

(80) Answer : (3)

Solution:

$$WAWB \Rightarrow pH = \frac{1}{2}(pk_w + pk_a - pk_b)$$

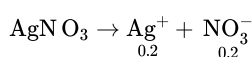
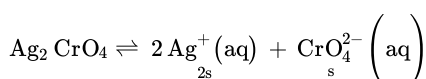
$$WASB \Rightarrow pH = \frac{1}{2}(pk_w + pk_a + \log C)$$

$$WBSA \Rightarrow pH = \frac{1}{2}(pk_w - pk_b - \log C)$$

$$SASB \Rightarrow pH = \frac{pk_w}{2}$$

(81) Answer : (3)

Solution:



$$K_{sp} = [Ag^+]^2 [CrO_4^{2-}]$$

$$1.1 \times 10^{-12} = (2s + 0.2)^2 (S)$$

$$\therefore (2s \ll 0.2)$$

$$1.1 \times 10^{-12} = (0.2)^2 (S)$$

$$\frac{1.1}{0.04} \times 10^{-12} = S$$

$$2.75 \times 10^{-11} = S$$

(82) Answer : (3)

Solution:

Order of negative electron gain enthalpy of halogens is.

Cl > F > Br > I

(83) Answer : (3)

Solution:

Un = 1

Nil = 0

enn = 9

Hence symbol $\Rightarrow$ Une

(84) Answer : (3)

Solution:

$$\Lambda_m = \frac{K \times 1000}{C}$$

$$= \frac{1.25 \times 10^{-3}}{0.2} \times 1000$$

$$= 6.25 \text{ S cm}^2 \text{ mol}^{-1}$$

(85) Answer : (2)

Solution:

$$n \times n_r = \frac{q}{F}$$

$$\frac{M}{M_2} \times n_F = \frac{4.825 \times 100}{96500}$$

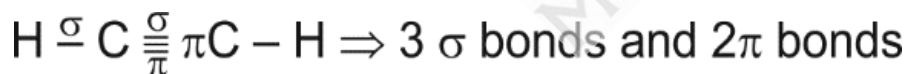
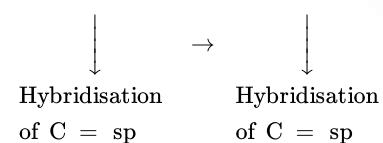
$$\frac{M}{\text{Eq. wt}} = 0.005$$

$$\frac{0.15}{0.005} = \text{Eq. wt}$$

$$30 = \text{Eq. wt}$$

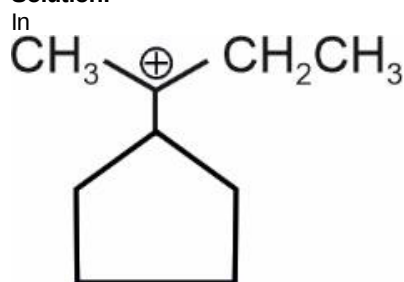
(86) Answer : (2)

Solution:



(87) Answer : (2)

Solution:



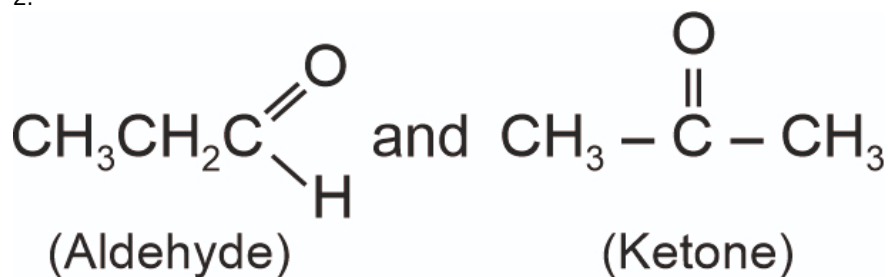
6, C – H bond are involve in hyperconjugation

(88) Answer : (4)

Solution:

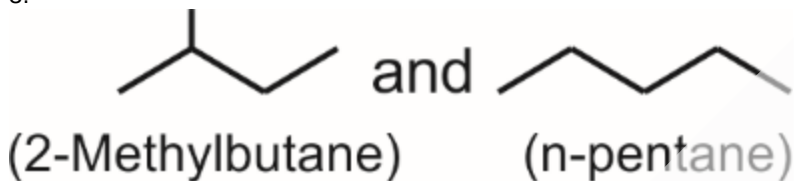
1. $\text{H}_3\text{C} - \text{O} - \text{CH}_2\text{CH}_3$ does not show metamerism

2.



are functional group isomers.

3.



are chain isomers

4.



These are not isomer as they do not have same molecular formula.

(89) Answer : (3)

Solution:

SnO_2 is an amphoteric oxide

(90) Answer : (2)

Solution:



There are two 3-centre 2-electrons bonds

BIOLOGY

(91) Answer : (2)

Solution:

Of the incident solar radiation less than 50% of it is PAR. Plants capture only 2-10% of the PAR and this small amount of energy sustains the entire living world.

(92) Answer : (4)

Solution:

Removal of carbon dioxide is observed during the process of Krebs' cycle, link reaction and alcoholic fermentation.

(93) Answer : (3)

Solution:

In chemiosmosis, the primary acceptor of electron which is located towards the outer side of the membrane transfers its electron not to an electron carrier but to an H-carrier.

Hence, this molecule removes a proton from the stroma while transporting an electron.
The NADP reductase enzyme is located on the stroma side of the thylakoid membrane.

(94) Answer : (2)

Solution:

Sorghum is a C_4 plant. The first stable product of CO_2 fixation is oxaloacetic acid (4C compound).

(95) Answer : (3)

Solution:

Net gain of ATP is two for a molecule of glucose converting into two molecules of pyruvic acid via glycolysis.

(96) Answer : (4)

Solution:

Vertical distribution of different species occupying different levels is called stratification.

(97) Answer : (4)

Solution:

Reproduction is shown by all living organism except a few which are either sterile or infertile like mule, worker-bees, etc.

(98) Answer : (1)

Solution:

Bacteria are the sole members of kingdom Monera. They have cell wall made up of peptidoglycan except in archaeobacteria and mycoplasma.

(99) Answer : (4)

Solution:

Rhizopus is an example of Phycomycetes in which sexual reproduction is observed.

(100) Answer : (4)

Solution:

Gamopetalous condition is represented in the floral formula as $C_{(n)}$.

(101) Answer : (3)

Solution:

The given floral characters are true for the members of family Brassicaceae.

The floral formula of brassicaceae family is $Ebr \oplus or \% \text{ } \text{ } K_{2+2} C_{\times 4} A_{2+4} \underline{G}_{(2)}$

(102) Answer : (3)

Solution:

The outer covering of endosperm in maize seeds separates the embryo by a proteinaceous layer called aleurone layer.

Remains of nucellus persist in the seeds of some plants and is called perisperm.

(103) Answer : (4)

Solution:

In racemose type of inflorescence, the main axis continues to grow, as it does not get terminated into flower. The flowers are borne laterally in acropetal succession.

(104) Answer : (2)

Solution:

The endoplasmic reticulum, which is free of ribosomes is known as the smooth endoplasmic reticulum. It is involved in the process of detoxification of drugs.

(105) Answer : (3)

Solution:

Nucleolus is the site for ribosomal RNA (rRNA) synthesis. They are more numerous in cells that are actively involved in protein synthesis.

(106) Answer : (2)

Solution:

The squamous epithelium is made of a single thin layer of flattened cells with irregular boundaries. They are found in the wall of blood vessels and air sacs of lungs and are involved in functions like forming a diffusion boundary.

(107) Answer : (4)

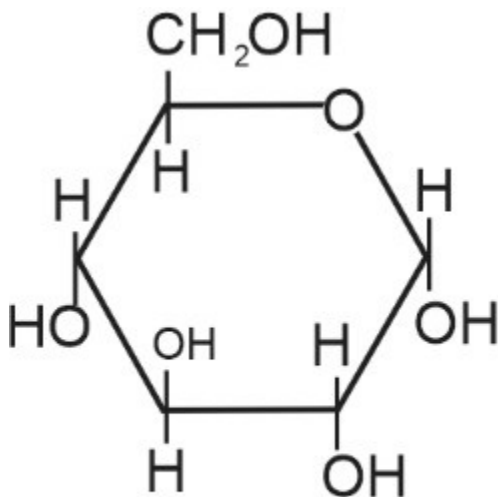
Solution:

Palmitic acid has 16 carbons including the carboxyl carbon. Arachidonic acid has 20 carbon atoms including the carboxyl carbon.

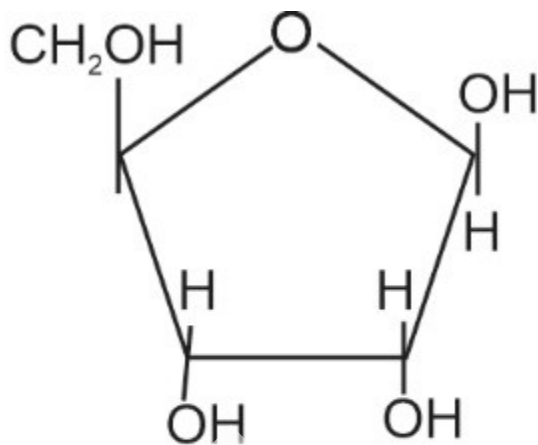
Cytidylic acid is a nucleotide. Glutamic acid is an amino acid.

(108) Answer : (3)

Solution:



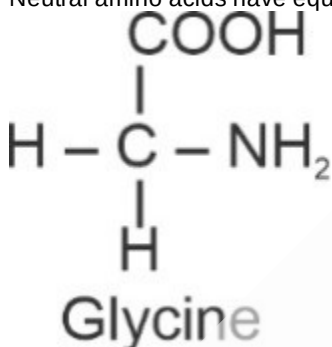
Both are monosaccharides.



(109) Answer : (2)

Solution:

Neutral amino acids have equal number of amino group(s) and carboxylic group(s).



Tyrosine is an aromatic amino acid.

(110) Answer : (2)

Solution:

Lecithin is a phospholipid. Glucagon, collagen, myoglobin and insulin are proteins and have peptide bonds; glycogen and inulin are polysaccharides and have glycosidic bonds in their structures.

(111) Answer : (2)

Solution:

Zinc is a co-factor for the proteolytic enzyme carboxypeptidase. Haem is the prosthetic group for peroxidase and catalase.

(112) Answer : (2)

Solution:

The bone marrow in some bones is the site of production of blood cells.

(113) Answer : (3)

Solution:

The neural organisation is very simple in lower invertebrates. For example, in *Hydra*, it is composed of a network of neurons called nerve net, lacking a brain. The neural system is very-well developed in vertebrates

(114) Answer : (3)

Solution:

Transformation - Introduction of foreign DNA in the host bacterium.

Spooling - Method of obtaining DNA in a form of spool over a glass rod.

Elution - Extraction of DNA out of the agarose gel.

Genome - Complete set of genetic information in an organism.

(115) Answer : (4)

Solution:

ori is a sequence from where replication starts and any piece of DNA when linked to this sequence can be made to replicate within the host cells. This sequence is also responsible for controlling the copy number of the linked DNA. So, if one wants to recover many copies of the target DNA, it should be cloned in a vector whose origin supports high copy number.

Bigger vector molecules are not desirable as they may break down during the process of purification.

(116) Answer : (1)

Solution:

G₁ phase or the post mitotic phase is involved in the duplication of majority of cell organelles.

(117) Answer : (2)

Solution:

Separation of chromatids is a common event between mitosis and meiosis II.

(118) Answer : (2)

Solution:

Primary cell wall tends to diminish with maturity.

(119) Answer : (3)

Solution:

Diakinesis is the stage which shows transition to metaphase I and it is also associated with terminalisation of chiasmata.

(120) Answer : (2)

Solution:

rRNAs play structural and catalytic role during translation.

(121) Answer : (4)

Solution:

Marshall Nirenberg's cell-free system for protein synthesis helped the genetic code to be deciphered.

(122) Answer : (3)

Solution:

Severo Ochoa enzyme	It is polynucleotide phosphorylase enzyme which was helpful in polymerising RNA with defined sequences.
Helicase	Facilitate ATP dependent unwinding of double helix of DNA
Guanylyl transferase	Catalyses addition of methyl guanosine triphosphate at 5'-end of primary transcript
DNA ligase	Join the two discontinuously synthesised fragments of DNA

(123) Answer : (2)

Solution:

Radial vascular bundles are found in roots of angiosperms.

(124) Answer : (3)

Solution:

Suberin is the waxy substance deposited in the radial and tangential wall of endodermal cells of both monocot and dicot roots.

(125) Answer : (4)

Solution:

DNA fingerprinting involves identifying differences in some specific regions in DNA sequence called repetitive DNA because in these sequences, a small stretch of DNA is repeated many times. These sequences show high degree of polymorphism and form basis of DNA fingerprinting.

(126) Answer : (1)

Solution:

Vessels are found to be absent in gymnosperms. Xylem parenchyma is the only living element of xylem. Both tracheid and xylem parenchyma are components of xylem tissue

(127) Answer : (2)

Solution:

According to Chargaff's rule, there is equimolar concentration of A = T and G = C in dsDNA.

G = C $\rightarrow$ 46 + 46 $\Rightarrow$ 92%
 Thus, A + T amount is 8%
 Therefore, 4% A and 4%T
 T = 4% and C = 46%

(128) Answer : (1)

Solution:

Tapetum is the innermost wall layer of microsporangium and is responsible for nourishing the developing pollen grains.

(129) Answer : (3)

Solution:

Cells proximal to the meristematic zone towards the stem base are in elongation zone.

Cells attain their maximal size in the phase of maturation.

(130) Answer : (2)

Solution:

Relative growth rate can be calculated by the formula:

$$\frac{\text{Growth per unit time}}{\text{Initial size}} \times 100$$

Hence, for leaf A: $\frac{20}{10} \times 100 = 200\%$

For leaf B: $\frac{10}{25} \times 100 = 40\%$

(131) Answer : (4)

Solution:

During gel electrophoresis, the DNA fragments separate (resolve) according to their size through the sieving effect provided by the agarose gel.

DNA molecules are negatively charged.

(132) Answer : (2)

Solution:

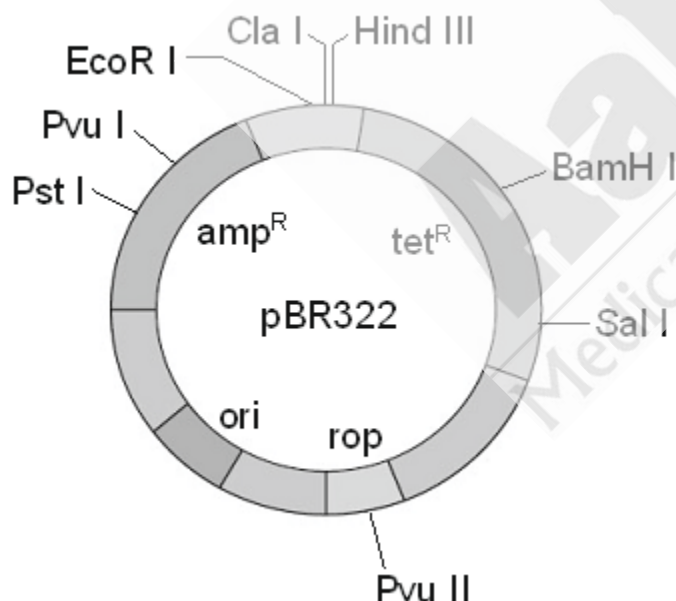
During the denaturation step of polymerase chain reaction, the hydrogen bonds present between the bases in the two strands of DNA are broken.

Peptide bonds are present in proteins.

Glycosidic bonds are present in carbohydrates.

(133) Answer : (3)

Solution:



(134) Answer : (4)

Solution:

The hypothalamus lies at the base of the thalamus. It contains a number of centres which control body temperature, urge for eating and drinking. It also contains several groups of neurosecretory cells, which secrete hormones called hypothalamic hormones.

The cerebrum wraps around a structure called thalamus, which is a major coordinating centre for sensory and motor signalling.

(135) Answer : (4)

Solution:

Each lobule of testis contains one to three highly coiled seminiferous tubules. Because each testis has 250 testicular lobules, so number of seminiferous tubules in each testis is about 250 - 750.

(136) Answer : (4)

Solution:

Vagina is not a part of the external female genitalia.

(137) Answer : (3)

Solution:

Olfactory lobes, optic lobes and cerebral hemispheres are paired structures present in the brain of frog whereas diencephalon is an unpaired structure.

(138) Answer : (1)

Solution:

In human females, each ovary is 2 – 4 cm in length and is connected to the pelvic wall and uterus by ligaments. It is covered by a thin epithelium which encloses the ovarian stroma. The stroma is divided into two zones – a peripheral cortex and an inner medulla.

Nephrons are the structural and functional unit of kidneys. Each fallopian tube is 10-12 cm in length.

(139) Answer : (2)

Solution:

In neural system of cockroaches, three ganglia lie in the thorax region called thoracic ganglia and six ganglia lie in the abdomen called abdominal ganglia.

(140) Answer : (2)

Solution:

Secretions of the male sex accessory glands constitute the seminal plasma which is rich in fructose, calcium and certain enzymes. Secretions of bulbourethral glands is responsible for lubrication of penis.

(141) Answer : (1)

Solution:

Both cartilaginous and bony fishes have streamlined body. Bony fishes have air bladder and usually exhibit external fertilisation. In cartilaginous fishes, teeth are modified placoid scales which are backwardly directed. Fertilisation is internal in them.

(142) Answer : (2)

Solution:

The theory of special creation has three connotations. One of the connotation is that, the diversity on the Earth was always the same since creation and will be the same in future.

(143) Answer : (3)

Solution:

There was no atmosphere on the early Earth. Water vapour, methane, carbon dioxide and ammonia released from molten mass covered the surface.

(144) Answer : (4)

Solution:

Osmoreceptors in the body are activated by changes in blood volume. ADH facilitates water reabsorption from latter parts of the renal tubule, thereby preventing diuresis. ADH can cause an increase in blood pressure which further increases the glomerular blood flow and thereby GFR.

(145) Answer : (4)

Solution:

Sea walnuts/comb jellies are exclusively marine, radially symmetrical, diploblastic organisms with tissue level of organisation.

(146) Answer : (4)

Solution:

Kidneys are situated between the levels of last (12th) thoracic and third lumbar vertebra close to the dorsal inner wall of the abdominal cavity. Each kidney of an adult human measures 10 – 12 cm in length, 5 – 7 cm in width and 2 – 3 cm in thickness.

(147) Answer : (2)

Solution:

Genetic modification has made crops more tolerant to abiotic stresses (cold, drought, salt, heat), reduced reliance on chemical pesticides (pest-resistant crops), helped to reduce post-harvest losses and increased efficiency of mineral usage by plants.

(148) Answer : (3)

Solution:

Fossils are remains of hard parts of life forms found in rocks. A study of fossils in different sedimentary layers indicates the geological period in which they existed.

(149) Answer : (2)

Solution:

Ctenoplane — Only sexual reproduction

Liver fluke — Internal fertilisation

Pennatula — Tissue level of organisation

Sea lily — Organ-system level of organisation

(150) Answer : (4)

Solution:

Very low concentration of DNA can be detected by amplification of nucleic acid sample by PCR with the help of *Taq* polymerase.

Denaturation step of PCR is performed at 94°C. Uracil is present in RNA.

(151) Answer : (4)

Solution:

Erythrocytes or RBCs are the most abundant of all the cells in blood. RBCs have red-colour, iron containing complex protein called haemoglobin. These molecules play a significant role in the transport of gases. Neutrophils and monocytes are phagocytic cells. Eosinophils resist infections and are also associated with allergic reactions. Platelets are involved in the coagulation or clotting of blood.

(152) Answer : (2)

Solution:

Arthritis is inflammation of joints. Gout is inflammation of joints due to accumulation of uric acid crystals. Myasthenia gravis is an auto-immune disorder, affecting neuro-muscular junction or motor-end plate.

(153) Answer : (1)

Solution:

Muscles have special properties like excitability, contractibility, extensibility and elasticity. Neurons are excitable cells.

(154) Answer : (4)

Solution:

In an ECG, Q-wave represents the initial phase of ventricular depolarisation.

Negative deflection : When the wave of depolarisation moves away from a specific electrode

(155) Answer : (4)

Solution:

When a host is exposed to antigens, which may be in the form of living or dead microbes or other proteins, antibodies are produced in the host body. This type of immunity is called active immunity.

When ready-made antibodies are directly given against any foreign agent, it is called passive immunisation. E.g.- Colostrum (IgA), ATS

(156) Answer : (2)

Solution:

In the given figure, A is thalamus, B is endocarp and C is mesocarp.

(157) Answer : (2)

Solution:

Epiphydrophily is observed in *Vallisneria* and hypophydrophily is observed in *Zostera*.

(158) Answer : (4)

Solution:

Feminine development, such as, gynaecomastia is the characteristic of Klinefelter's syndrome.

(159) Answer : (1)

Solution:

Mendelian disorders are mainly determined by alteration or mutation in the single gene.

(160) Answer : (2)

Solution:

The possible genotypes of the parents can be

Father – $I^A i$; Mother – $I^B i$

Father – $I^A I^A$; Mother – $I^B i$

(161) Answer : (1)

Solution:

Mendel's law of independent assortment does not hold true for the genes that are located on the same chromosome because closely located genes assort together.

(162) Answer : (4)

Solution:

Archegoniates are bryophytes, pteridophytes and gymnosperms. Bryophytes are non-vascular.

(163) Answer : (1)

Solution:

Brown algae have cell wall made up of cellulose which is usually covered on the outside by a gelatinous coating of algin e.g. *Fucus*, *Sargassum*.

(164) Answer : (3)

Solution:

Streptokinase is produced by the bacterium *Streptococcus* and is modified by genetic engineering. It is used as a 'clot buster' for removing clots from the blood vessels of patients who have undergone myocardial infarction leading to heart attack.

(165) Answer : (4)

Solution:

The type of interaction in which one species is benefitted and other is neither harmed nor benefitted, is commensalism. Barnacles growing on the back of a whale show commensalism.

(166) Answer : (2)

Solution:

Tropical environments are less seasonal, relatively more constant and predictable. Such constant and favourable environment has helped tropical organisms to gain more niche specialisation and lead to greater species diversity.

(167) Answer : (4)

Solution:

Anthropogenic extinction is causing a sixth mass extinction of species. It is 100-1000 times faster than the rate of natural extinctions.

(168) Answer : (3)

Solution:

In *Ulothrix*, gametes are flagellated and similar in size. Cell wall is rigid made up of an inner layer of cellulose and an outer layer of pectose.

(169) Answer : (3)

Solution:

Primary treatment is a physical process which involves removal of large and small particles from sewage through filtration and sedimentation.

(170) Answer : (1)

Solution:

In field experiment, when all the starfish (*Pisaster*) were removed experimentally from an enclosed intertidal area more than 10 species of invertebrates become extinct within a year because of interspecific competition.

(171) Answer : (4)

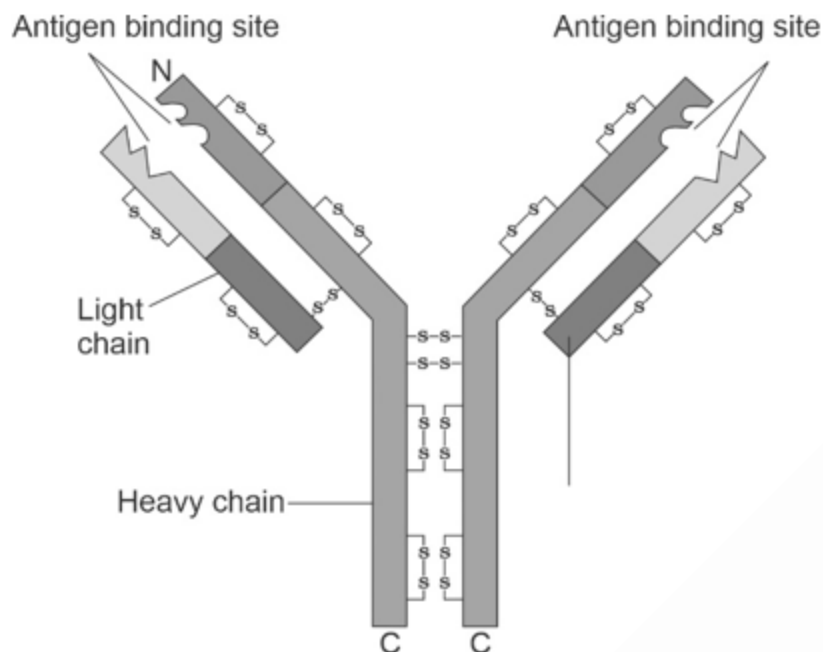
Solution:

The adrenal gland is composed of two types of tissues. The centrally located tissue is called the adrenal medulla, and outside this lies the adrenal cortex.

The adrenal cortex can be divided into three layers, called zona reticularis (innermost), zona fasciculata (middle layer) and zona glomerulosa (outermost).

(172) Answer : (2)

Solution:



(173) Answer : (2)

Solution:

Cell-mediated immunity is responsible for graft rejection and it is mediated by T-lymphocytes. The T-cells do not secrete antibodies themselves but help B-cells to produce them. T-cells require both bone marrow and thymus for their maturation.

(174) Answer : (3)

Solution:

- Malarial parasites reproduce asexually in liver cells.
- Malarial parasites reproduce asexually in RBCs of human, bursting the RBCs and causing chills and high fever due to release of haemozoin.
- Fertilization of gametes and development of zygote occur in the mosquito's gut.
- Mature infective stage, i.e., sporozoites, escape from mosquito's gut and migrate to the mosquito's salivary glands.

(175) Answer : (1)

Solution:

AI, GIFT and IUI involve *in vivo* fertilization whereas ICSI and ZIFT involve *in vitro* fertilization.

Intra Cytoplasmic Sperm Injection (ICSI) is a specialised procedure to form an embryo in the laboratory in which a sperm is directly injected into the ovum. *In vitro* fertilization (IVF) is fertilization outside the body of the female in almost similar conditions as that in the body.

(176) Answer : (2)

Solution:

Periodic abstinence is one of the natural methods of preventing conception in which the couples avoid or abstain from coitus from day 10 to 17 of the menstrual cycle when ovulation could be expected. As chances of fertilisation are very high during this period, it is also called the fertile period.

(177) Answer : (4)

Solution:

Mature insulin lacks C-peptide.

Pro-insulin is the inactive form of mature insulin.

(178) Answer : (4)

Solution:

Haemoglobin is a red-coloured iron containing complex pigment present in the RBCs. O_2 can bind with haemoglobin in reversible manner to form oxyhaemoglobin. Each haemoglobin molecule can carry a maximum of four molecules of O_2 .

CO_2 combines haemoglobin to form carbamino-haemoglobin.

(179) Answer : (3)

Solution:

An ideal contraceptive should in no way interfere with the sexual drive, desire and/or the sexual act of the users.

Contraceptives are not regular requirements for the maintenance of reproductive health.

(180) Answer : (3)

Solution:

If the levels of CO_2 increases in blood, then the rate of respiration will increase. Increased levels of CO_2 in blood will make the blood acidic. Neural signals from the pneumotaxic centre can reduce the duration of inspiration and there by alter the respiratory rate

